

João Carlos Jachenski Junior

Machine Learning and Data Applied to Systems

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SUMMARY

Electrical Engineering student at the Federal University of Paraná (UFPR), with a technical degree in Computer Science from TECPUC. Research fellow at Lactec Institutes, where I build machine learning models for photovoltaic and power systems, from soiling detection on PV modules to consumption and load forecasting. Co-author of 3 journal articles and 6 software programs registered with INPI, Brazil's patent office. I maintain open-source Python projects with automated tests and CI.

EXPERIENCE

Research Fellow · Lactec Institutes, CFG Department, Curitiba — 2025 – present

- Machine learning models for photovoltaic monitoring: soiling status detection and cleaning alerts from inverter and weather data.
- Multi-horizon residential electricity consumption forecasting and load profile estimation for center-pivot irrigation systems.
- End-to-end pipeline, from processing weather and sensor data to deployable embedded models (Edge AI).
- Results: 6 software programs registered with INPI and 3 journal articles, 2 of them in *Energies* (MDPI, JCR-indexed).

Research Project Member · Rede Agrointeligente (EMBRAPII) — 2025 – present

- Multi-institution project on energy management for irrigated agriculture, combining renewable generation and storage to cut costs and emissions on farms. Led by Bruno Knevitz Hammerschmitt.

Team Member, Accumulator · UFPR Fórmula (Formula SAE Electric) — 2024 – 2025

- Worked on the accumulator (high-voltage battery pack) of the electric competition car.

Sales Team Member · EMJEL, Electrical Engineering Junior Enterprise (UFPR) — 2024 – 2025

- Client prospecting and negotiation of the junior enterprise's electrical engineering projects.

PROJECTS

prosumer: Python library for day-ahead forecasting of PV generation and household consumption, the basis of my undergraduate thesis. Every default was chosen by a benchmark of 57 models on real plants and houses. Python, pandas, LightGBM, time-series foundation models, FastAPI, React

News Radio: pipeline that turns the day's economy news into a two-voice podcast episode, with a local LLM and local text-to-speech. 221 tests that run with no GPU, API key or network. Python, asyncio, Ollama, Kokoro TTS, pytest, GitHub Actions

academic-search: searches OpenAlex, Scopus, IEEE Xplore and Semantic Scholar at once, with deduplication, semantic-similarity ranking and flagging of retracted papers. Python, httpx, sentence-transformers, SQLAlchemy, Streamlit

Loom: cross-domain recommender (anime, books, music, movies and TV) as a local desktop app, with text embeddings and collaborative filtering over 25 million ratings. Python, FastAPI, SQLite, embeddings, collaborative filtering

EDUCATION

B.Sc. in Electrical Engineering (in progress) · Federal University of Paraná (UFPR) — 2023 – present

Computer Science Technician, integrated with high school · TECPUC, Curitiba — 2020 – 2022

PUBLICATIONS

1. Hammerschmitt, B. K.; **Jachenski Junior, J. C.**; Mario, L.; Tonolo, E. A.; Fonseca, P. H.; Silva, R. M.; Menezes, N. P. *Soiling Status Detection in Photovoltaic Energy Systems Using Machine Learning and Weather Data for Cleaning Alerts*. Energies, v. 19, p. 1964, 2026.
2. Luz, T. J.; Leone, A. S.; Tonolo, E. A.; Rambo, M. V. H.; **Jachenski, J. C.**; Mantovan, F. M.; Rodolfo, T. A.; Schneider, P. S.; Hammerschmitt, B. K. *Sustainable Hybrid Off-Grid Energy Systems for Rural Electrification in the Amazon Region*. Energies, v. 19, p. 5104, 2026.
3. **Jachenski Junior, J. C.**; Rambo, M. V. H.; Leone, A. S.; Lopes, R. F. L.; Iantorno, L. M.; Riella, R. J.; Hammerschmitt, B. K. *A Comparison of State-of-the-Art Time Series Classification for Demand Estimation in Prosumers*. IOSR Journal of Business and Management, v. 27, p. 68–73, 2025.

REGISTERED SOFTWARE (INPI, BRAZIL)

BR512026006223-3	PivotEnergy: load profile estimation for center-pivot irrigation systems	2026
BR512026006715-4	SciKeywordMap: keyword neural network (UFPR)	2026
BR512026003168-0	Online soiling monitoring and alert system for photovoltaic modules	2026
BR512026000942-1	Multiclass soiling monitoring and alert system for photovoltaic systems	2026
BR512026000941-3	Probabilistic framework for identifying the operating state of photovoltaic systems, with a cleaning alert system	2025
BR512025006319-9	Multiscale system for multi-horizon residential electricity consumption forecasting	2025

SKILLS

Machine learning	scikit-learn, LightGBM, TensorFlow/Keras, time-series forecasting and classification, embedded models (TFLite)
Programming	Python (pandas, NumPy, FastAPI, SQLAlchemy, pytest), SQL/SQLite, JavaScript/TypeScript, React, HTML, C++
Tools	Git, GitHub Actions (CI), Excel, AutoCAD, Cadence
Domain	Photovoltaic systems, power and energy systems, microelectronics design

LANGUAGES

Portuguese: native
English: advanced
German: intermediate
French: basic

COURSES

English: INFLUX English School, 224 h (2016 – 2019)
German: Deutsch Haus (2023 – present)